

A2: TEC Heating and Cooling Analysis

Assessment code: A2

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Repository: <https://github.com/kapilnaidu1/phys39-naidu-cohen>, branch main

1. Steady-state temperature versus signed PWM

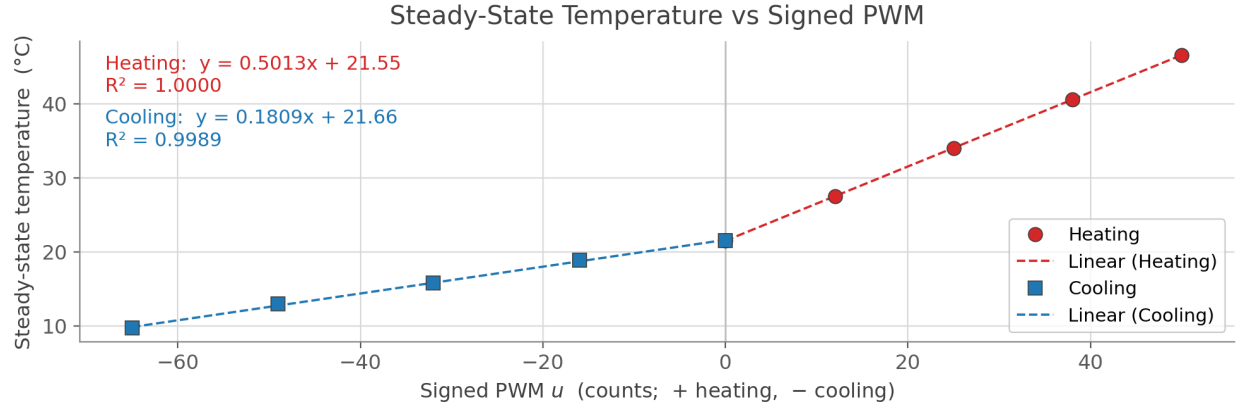


Figure 1. Steady-state plate temperature against the signed PWM command u , positive for heating and negative for cooling. Ten points, five per branch, about the zero-command temperature $T_0 = 21.54$ °C. Each trendline is fitted only over its own branch.

Steady-state criterion: after each step the trace was watched for about three driven time constants ($\tau \approx 70$ s) plus one further minute, and the point was accepted once its net drift over that minute was no larger than the measured noise of 0.16 °C peak to peak. Points that had not fully arrived are reported as the asymptote of an exponential fit, and data/module_04/steady_state.csv records which.

2. Slopes, units, fit ranges and ratio

Branch	Susceptibility dT/du	Fit range	R^2	Intercept
Heating	$m_h = 0.5013$ °C per PWM count	$u = 0$ to $+50$	1.0000	21.55 °C
Cooling	$m_c = 0.1809$ °C per PWM count	$u = -65$ to 0	0.9989	21.66 °C

Both branches are linear with no visible curvature, so neither slope is a local estimate. $r = m_h / m_c = 2.771$. The intercepts differ by 0.11 °C and the cooling one sits 0.12 °C above T_0 , consistent with the room drifting over a two-hour session.

3. Derivation

3.1 PWM current averages

Over one PWM period τ the bridge delivers current I for a time $D\tau$ and zero for the remaining $(1-D)\tau$, with duty $D = |u| / 255$. Both integrands vanish outside the on-interval, so

$$\begin{aligned} \langle I \rangle &= (1/\tau) \int I dt = (1/\tau) \cdot I \cdot D\tau = D I \\ \langle I^2 \rangle &= (1/\tau) \int I^2 dt = (1/\tau) \cdot I^2 \cdot D\tau = D I^2 \end{aligned}$$

$\langle I^2 \rangle$ is not $\langle I \rangle^2 = D^2 I^2$: squaring and averaging do not commute, and for a two-level waveform the two differ by a factor $1/D$. Peltier transport follows $\langle I \rangle$ and Joule heating follows $\langle I^2 \rangle$, and under PWM both are linear in D , so the susceptibility dT/du is constant and the graph is straight. Had the Joule term gone as D^2 the branches would curve visibly; they do not. With a DAC supplying a steady current of variable amplitude instead, $\langle I^2 \rangle = \langle I \rangle^2$ would hold, so the distinction is a property of PWM.

3.2 Steady-state energy balance

With C the heat capacity of the object and G the conductance between it and the room, $C dT/dt = Q_{TEC} - G(T - T_0)$. At steady state $dT/dt = 0$, so the individual flows are not zero but their sum is, and $G(T - T_0) = Q_{TEC}$. With signed duty $d = u / 255$, Peltier transport reverses with the current and Joule heating does not, so $Q_{TEC} = d Q_p + |d| Q_j$. On the heating branch $d > 0$ and both terms deliver heat into the object; on the cooling branch $d < 0$ and Peltier removes heat while Joule still adds it. Both derivatives below are positive while $Q_p > Q_j$, and the heating one is the larger. Hence

$$\begin{aligned} T_h - T_0 &= d (Q_p + Q_j) / G \rightarrow dT_h/dd = (Q_p + Q_j) / G \\ T_c - T_0 &= d (Q_p - Q_j) / G \rightarrow dT_c/dd = (Q_p - Q_j) / G \end{aligned}$$

3.3 Slope ratio and numerical result

Since $d = u / 255$, each measured slope carries the same factor $1/255$, and that factor and G are common to both branches, so both cancel from the ratio:

$$r = m_h/m_c = (Q_p + Q_j) / (Q_p - Q_j) \Rightarrow Q_j/Q_p = (r - 1)/(r + 1) \quad \text{check: } r = 2 \text{ gives exactly } 1/3$$

$$r = 2.771 \text{ gives } Q_j/Q_p = 1.771 / 3.771 = 0.470$$

Joule heat reaching the object face is about 47% of the Peltier pumping. G cancels from r, which matters because G was never measured.

4. Laird data sheet

Source: Laird CP14-127-045-L2-W4.5, MFG part 58910-501, SPECIFICATIONS table, hot side temperature 27.0 °C column.

Quantity	Value	Meaning, and the condition attached
R_M	1.50 Ω	Ohmic resistance of the 127 couples in series, quoted at $T_h = 27$ °C and rising to 1.68 Ω at 50 °C
$I_{\max}$	8.6 A	The current giving the largest temperature difference, not a safety ceiling. Condition: I at $\Delta T_{\max}$
$Q_{c,\max}$	71.3 W	Most heat the cold face can absorb. Condition: $\Delta T = 0$, no difference to pump against
$\Delta T_{\max}$	70.5 °C	Largest sustainable temperature difference. Condition: $Q_c = 0$, no heat load

The last two sit at opposite ends of one trade-off: maximum pumping needs zero temperature difference, maximum difference needs zero load. At the data-sheet condition $\Delta T = 0$, so conduction through the module vanishes and the symmetric model puts half the Joule heat on each face:

$$Q_{j,\max} = \frac{1}{2} I_{\max}^2 R_M = \frac{1}{2} (8.6 \text{ A})^2 (1.50 \Omega) = 55.47 \text{ W}$$

$$Q_{p,\max} = Q_{c,\max} + Q_{j,\max} = 71.3 \text{ W} + 55.47 \text{ W} = 126.77 \text{ W}$$

$$r_{\text{Laird}} = (126.77 + 55.47) / (126.77 - 55.47) = 2.556$$

Cross-check. $V_{\max}/I_{\max} = 13.9 / 8.6 = 1.616 \Omega$, 8% above the quoted R_M , because at $\Delta T_{\max}$ the module develops a Seebeck back-EMF: $V_{\max} = I_{\max} R_M + S \Delta T_{\max}$ gives $S = 0.0142 \text{ V/K}$, a reasonable coefficient for 127 bismuth telluride couples. The right rows were read.

5. Comparison

	r	Q_j/Q_p	Condition
Measured, this apparatus	2.771	0.470	PWM drive, finite ΔT , current set by the supply and bridge
Laird, at $I_{\max}$	2.556	0.438	steady DC at 8.6 A, $\Delta T = 0$
ratio	1.084	1.073	agreement to 8%

The two agree to 8%, closer than the comparison warrants, since the data-sheet figures describe a condition this apparatus never reaches:

- **D = 1 does not mean I = $I_{\max}$** . Full duty only means the bridge is continuously on. The current is set by the 12 V supply and its limit, the H-bridge forward drop, wiring resistance and R_M , which together hold the bench current well below 8.6 A.
- **PWM, not steady DC**. The bench current is chopped; the data sheet assumes a steady one.
- **Finite ΔT** . The data-sheet relation is stated at $\Delta T = 0$, while the plate ran 25 °C above and 12 °C below ambient, where conduction through the module is absorbed into G.
- **Coefficients move with temperature**. R_M rises from 1.50 to 1.68 Ω between 27 and 50 °C, and one slope is fitted across each whole branch.

Direction of the difference. The measured ratio is the larger, yet running below $I_{\max}$ should *reduce* Q_j/Q_p , since that ratio scales as I. The opposite sign points to dissipation this model assigns to the object face that the data sheet does not carry, such as lead and contact resistance in series with the module.

6. Passive conduction

When the object is hotter than the room, heat flows out of it; when it is colder, heat flows in. Either way the flow drives the object back toward room temperature, so conduction opposes the TEC on both branches, which is why it enters as $-G(T - T_o)$ with a single positive G.

It cannot explain the unequal slopes. The same G appears in the denominator of both dT_h/dd and dT_c/dd , so it sets how steep the branches are, a larger G flattening both, but it cancels exactly from their ratio: a symmetric term cannot produce an asymmetric result. The asymmetry survives only in the numerators, where Q_j is added on the heating branch and subtracted on the cooling one.

7. Conclusion

Steady-state temperature is linear in signed PWM on both branches ($R^2 = 1.0000$ heating, 0.9989 cooling), which is what the PWM averages predict: Peltier transport follows $\langle I \rangle = DI$ and Joule heating follows $\langle I^2 \rangle = DI^2$, so both scale with duty and the susceptibility stays constant. Had $\langle I^2 \rangle$ gone as D^2 , the branches would curve. The slopes differ by $r = 2.771$ because reversing the current reverses Peltier pumping but not Joule heating, so the two add when heating and oppose when cooling, giving $Q_j/Q_p = (r - 1)/(r + 1) = 0.470$. Conduction enters both branches through the same G and cancels from the ratio, setting how steep the lines are but not making them unequal. The Laird maximum-current figures predict $r = 2.556$, within 8% of a condition this apparatus never reaches.